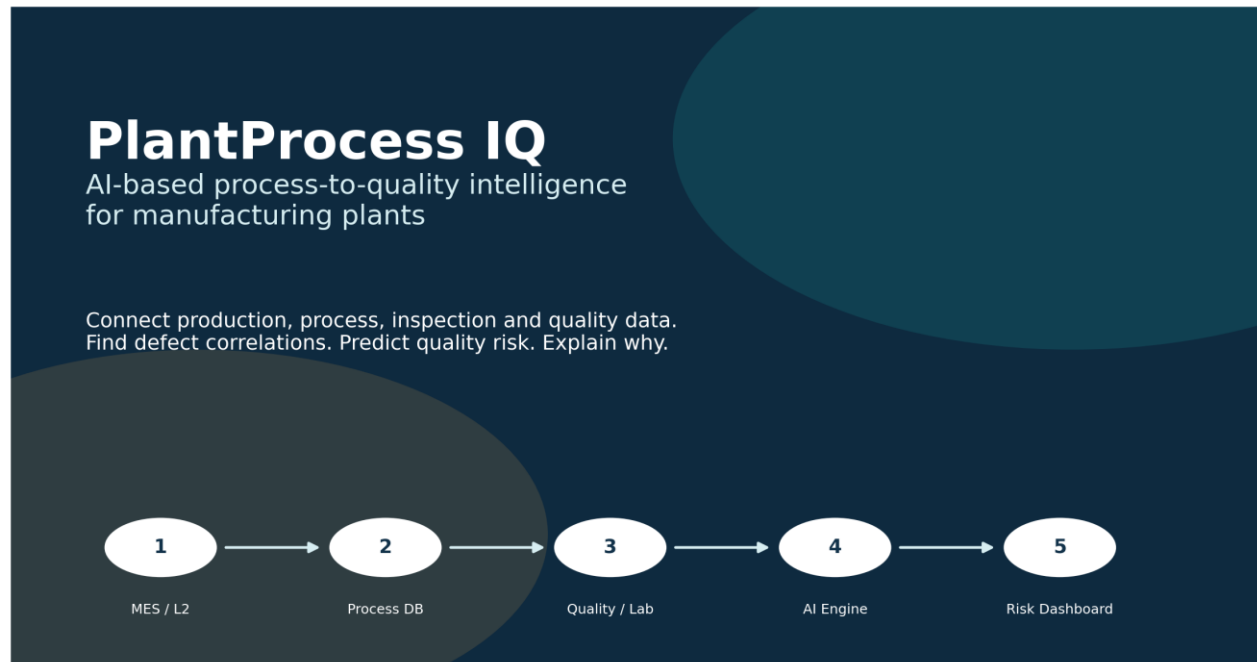


PlantProcess IQ

Customer Proposal: AI-Based Process-to-Quality Intelligence for Manufacturing



Prepared for: Karim Gamal Elsayed
Version 1.0 | 2026

1. Proposal Summary

PlantProcess IQ helps manufacturing companies connect production, process, inspection and quality data across the plant to understand why defects happen, predict quality risk earlier, and reduce manual investigation time. The solution is generic by design and can be configured for steel, aluminum, pharma, tire, paper, automotive, food, chemicals and other manufacturing domains.

Customer promise
Connect your plant data. Discover defect correlations. Predict quality risk. Explain suspected causes. Act earlier.

2. The Problem We Solve

From Data Silos to Quality Intelligence

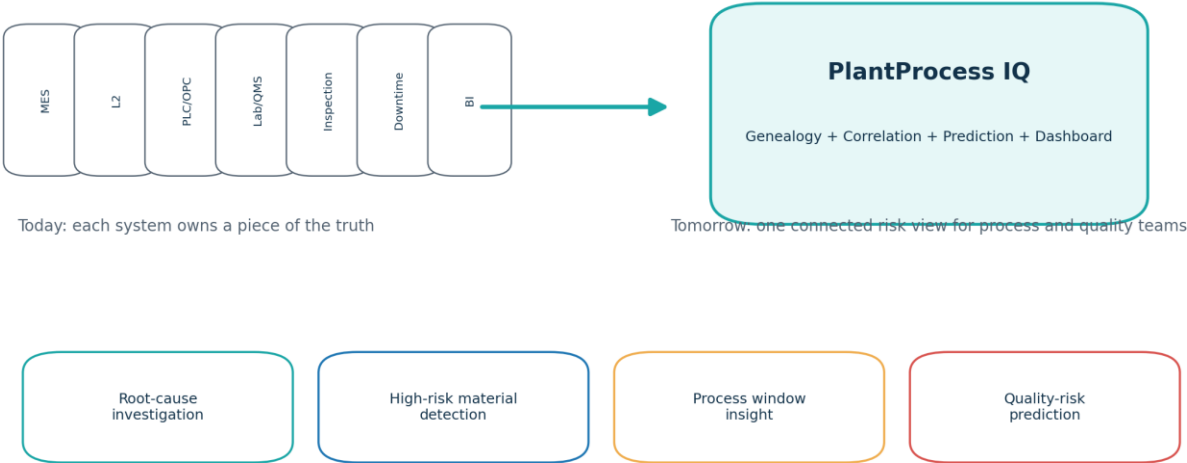


Figure 1 - Moving from fragmented data silos to connected quality intelligence.

- Plant data is spread across MES, Level 2, SCADA/PLC/OPC, historian, lab, inspection, downtime and ERP systems.
- Quality teams often investigate defects manually using Excel, screenshots, SQL queries and tribal process knowledge.
- Dashboards show what happened, but often do not explain which process parameters are linked to defects.
- Production teams need earlier warning when a material/batch is likely to create quality issues in the next process step.
- Management needs a repeatable method to learn from historical data and reduce scrap, rework, downgrade and claims.

3. What We Offer

Offer	Customer Outcome	Typical Duration
Data Readiness Assessment	Understand available data, gaps, quality, source systems and feasibility.	1-2 weeks
Quality Correlation Diagnostic	Identify which process parameters, equipment, route, shift or material conditions are linked to defects.	3-6 weeks
Quality Risk Prediction Pilot	Build an MVP model that predicts risk for a selected defect or quality KPI.	6-10 weeks
Dashboard & AI Explanation Pack	Provide dashboards, risk views, material	2-6 weeks

	genealogy and AI-assisted explanations.	
Production Deployment	Install on-prem or private cloud with connectors, security, reports and support.	3-6 months

4. Solution Overview

PlantProcess IQ - Generic Reference Architecture

A domain-neutral intelligence layer above MES, L2, historian, inspection, quality and ERP data.

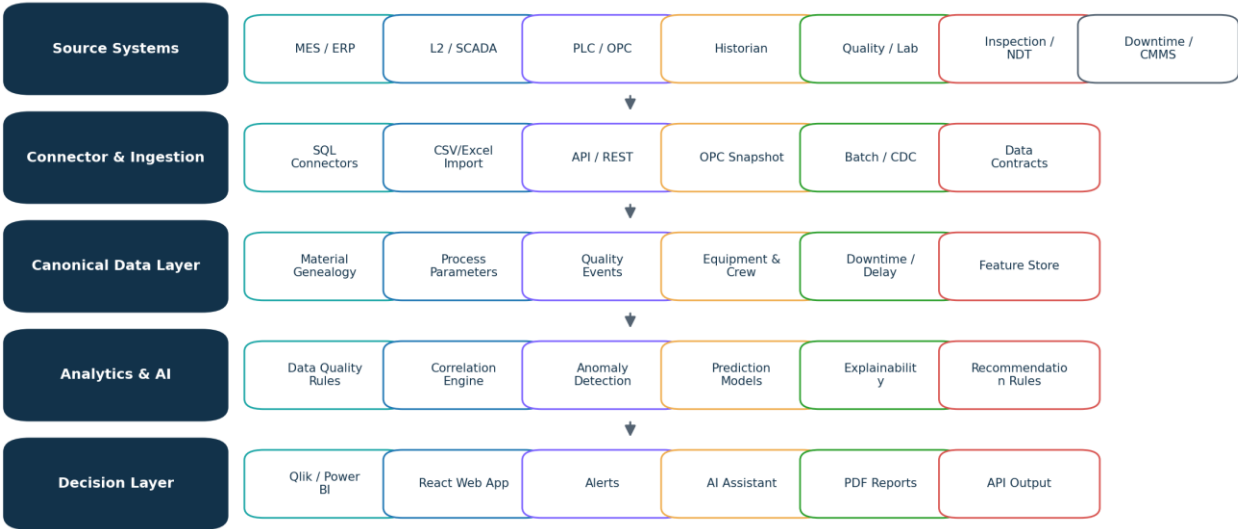
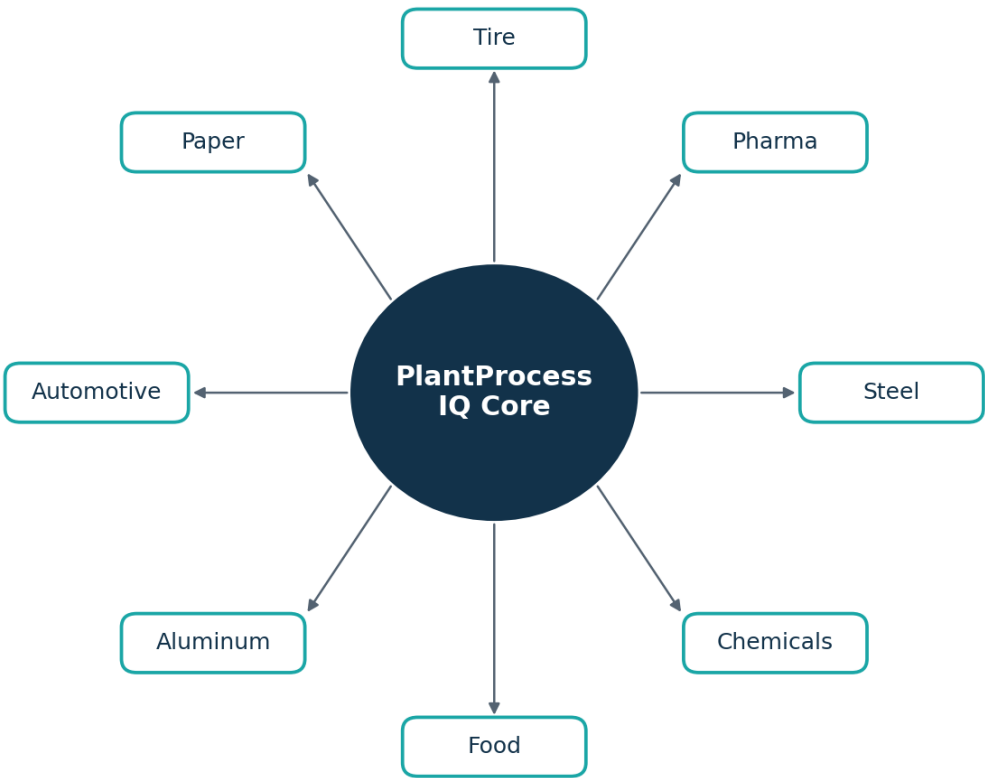


Figure 2 - High-level solution architecture.

Capability	Description
Data connectors	Connect to SQL Server, Oracle, PostgreSQL, CSV/Excel exports, MES/L2, inspection, quality and downtime sources.
Genealogy engine	Trace product/batch/material across process steps and link it to process and quality history.
Data quality checks	Find missing, duplicated, delayed or inconsistent production and quality records.
Correlation engine	Discover parameter ranges, equipment, routes or operating patterns linked to defects.
Prediction models	Score active or historical materials by quality-risk probability.
Dashboard and reports	Visualize quality KPIs, risk drivers, high-risk materials, process windows and investigation reports.

5. Generic Across Manufacturing Industries

One Generic Core - Many Manufacturing Domains



Reusable entities: batch / material / operation / equipment / parameter / quality event / prediction / recommendation

Figure 3 - The same platform core can be configured for multiple industries.

Industry	Example Data Sources	Example Quality Outcomes
Steel / Aluminum	EAF, LF, caster, rolling, inspection, quality lab	Cracks, inclusions, surface defects, profile/flatness, downgrade
Pharma	Batch records, lab results, equipment parameters, environmental monitoring	Out-of-specification, contamination risk, batch release delay
Tire	Mixing, curing, temperature/pressure, inspection	Uniformity issues, curing defects, dimensional defects
Paper	Machine speed, moisture, coating, web breaks, lab tests	Breaks, coating defects, moisture issues, roll defects
Automotive	Assembly parameters, torque, inspection, rework, supplier lot data	Rework, dimensional defects, warranty risk
Food / Chemicals	Batch process, temperature, pressure, recipe, lab/QA	Quality deviation, contamination risk, process instability

6. Example Dashboard Experience

Quality Risk Dashboard

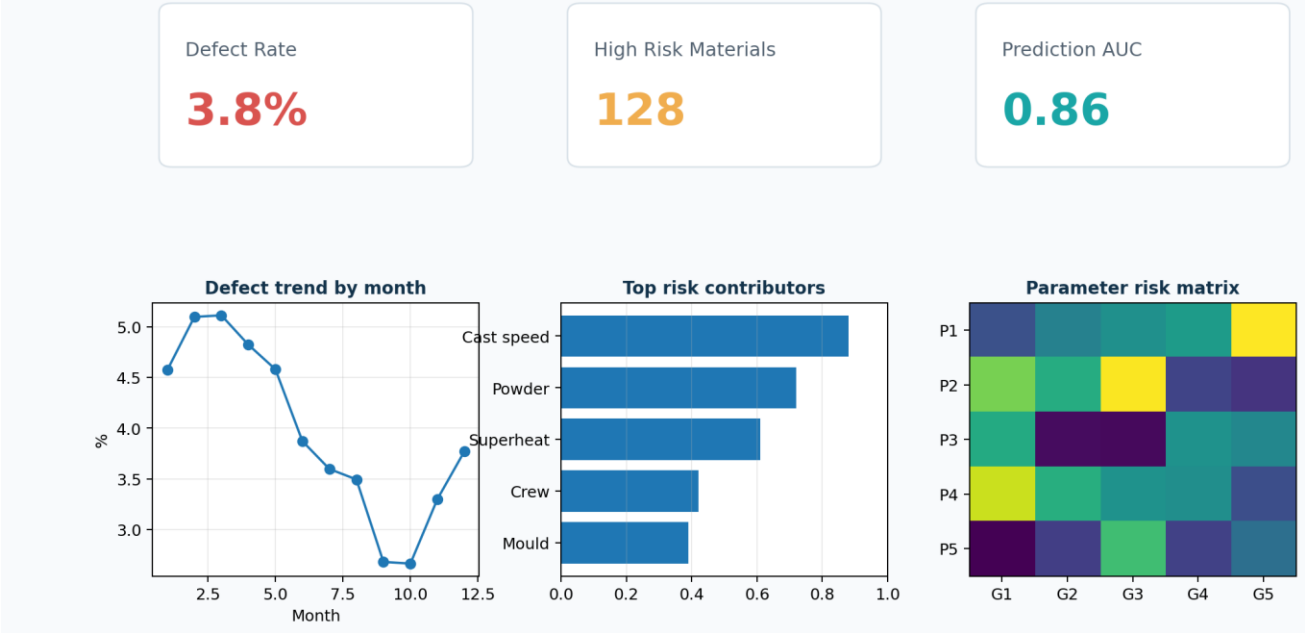


Figure 4 - Example quality-risk dashboard mockup.

- Quality overview: production volume, defect rate, rejection/downgrade and trend over time.
- Correlation analysis: defect rate by parameter range, equipment, route, grade, shift and material family.
- Material genealogy: search one product/batch and see its full process history.
- High-risk materials: list of products/batches with risk score and suspected contributors.
- AI explanation: plain-language answer to “why is this material high risk?”
- Recommended action: investigation or process-window suggestions validated by process experts.

7. Pilot Packages and Pricing Direction

Package	Scope	Output	Indicative Price
Data Readiness Assessment	Review data availability, sample tables, quality labels, genealogy feasibility.	Data readiness report + pilot scope.	EUR 3k-7k
Correlation Diagnostic Pilot	One process route + one quality topic using historical data.	Dashboard + correlation report + suspected contributors.	EUR 10k-25k
Prediction Model Pilot	Feature engineering + baseline ML model + risk dashboard.	Risk model + validation + dashboard + report.	EUR 25k-60k
Annual License / Support	Repeatable tool, connectors, dashboards, model updates and support.	On-prem or private-cloud product package.	EUR 50k-150k+/year

Final pricing depends on scope, number of data sources, data quality, deployment model, number of users, security requirements and support expectations.

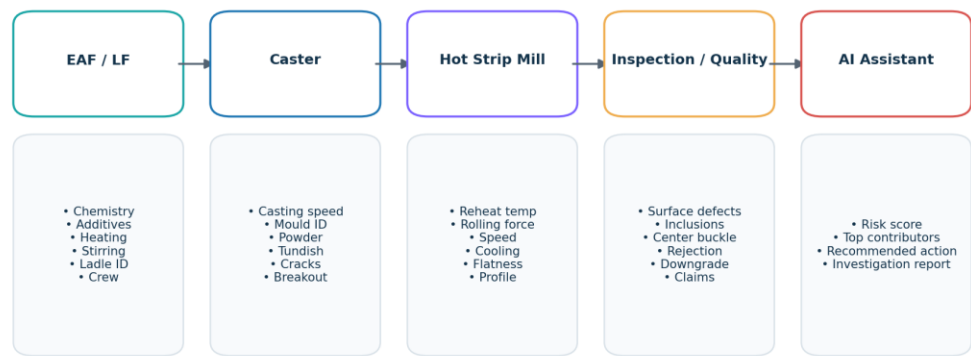
8. Why This Is Different from a Normal BI Dashboard

Normal Dashboard	PlantProcess IQ
------------------	-----------------

Shows KPIs after the fact.	Connects process history with quality outcomes and predicts risk.
Usually depends on manual interpretation.	Ranks suspected contributors and explains high-risk cases.
Often works per system or report.	Builds cross-system material/batch genealogy.
Shows defect count and trend.	Finds parameter windows and operating patterns linked to defects.
Designed mainly for reporting.	Designed for diagnosis, learning, prediction and prevention.

9. Steel Plant Example - End-to-End Quality Intelligence

Flat Steel Example: From EAF Parameters to HSM Quality Prediction



Outcome: detect risky combinations early, explain suspected contributors, and recommend prevention actions before expensive defects occur.

Figure 5 - Flat steel example from EAF/LF to Caster, HSM and Quality.

In a flat steel scenario, the system can combine EAF/LF chemistry, additives, heating and stirring time with caster speed, powder, mould, tundish and breakout/crack events, then link them to hot rolling parameters and final quality outcomes. The result is a risk model that can detect high-risk slabs or coils and recommend investigation or process adjustments before the defect becomes expensive.

10. Final Visual Sales Pages

Customer Pilot Delivery Flow

A paid pilot should produce business insight even before full product licensing.

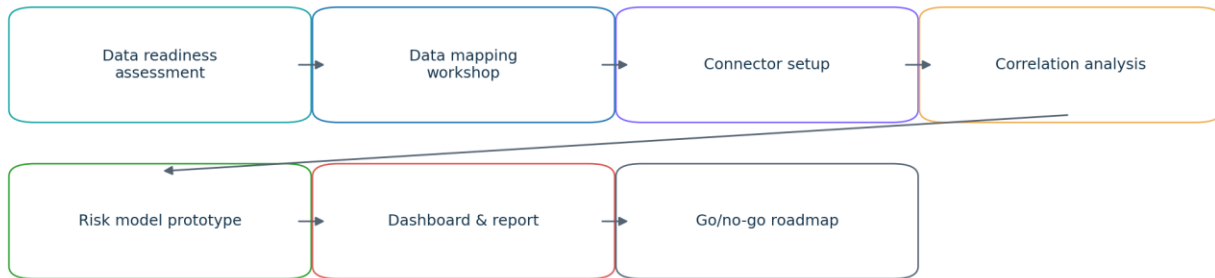


Figure 6 - Customer pilot delivery flow.

Page 1 - Start small, prove value fast

A pilot does not require replacing MES or connecting the whole plant. Start with one route, one defect family and one historical dataset. The first deliverable is business insight: what parameters appear linked to quality risk?

Page 2 - From historical learning to active prediction

Once historical correlation is proven, the same model can score active materials or batches before the next process step. This creates early warning for quality teams and process engineers.

Page 3 - Generic platform, industry-specific value

The same platform works across industries because the core model is generic: material/batch, operation, equipment, parameter, quality event and prediction. Each industry gets its own configuration pack.

11. Next Step

Recommended next step is a data readiness workshop. The workshop identifies available source systems, data quality, material genealogy feasibility, target quality topic, and the best pilot scope. After the workshop, the customer receives a fixed-scope proposal for the first correlation diagnostic pilot.

12. References

- Microsoft .NET support policy and release lifecycle: <https://dotnet.microsoft.com/en-us/platform/support/policy>
- ML.NET documentation: <https://learn.microsoft.com/en-us/dotnet/machine-learning/>
- scikit-learn documentation: <https://scikit-learn.org/>

- Qlik Sense analytic connections / Server-Side Extension: https://help.qlik.com/en-US/sense-developer/November2025/Content/Sense_Helpsites/AnalyticConnections.htm
- PostgreSQL declarative partitioning: <https://www.postgresql.org/docs/current/ddl-partitioning.html>
- PostgreSQL JSON/JSONB types: <https://www.postgresql.org/docs/current/datatype-json.html>
- ONNX Runtime documentation: <https://onnxruntime.ai/docs/>